Towards Romanian Sign Language Translation: A Broadcast News Corpus and Its Analysis

Anonymous ACL submission

Abstract

Romanian Sign Language (LSR) is used by an estimated 20,000–25,000 Deaf people in Romania, yet it remains almost absent from the sign language processing literature. The few existing LSR resources target *isolated* sign recognition, a formulation that does not scale to the open vocabulary of everyday communication. We describe the construction and analysis of a broadcast-derived LSR corpus assembled from publicly available news programmes of four Romanian television networks (TVR Iasi, Prima TV, Digi24 and Euronews Romania). The raw material comprises approximately 902 hours of footage, which we segment into 16,638 topically coherent news items paired with their Romanian transcripts. We characterise the corpus along dimensions that directly condition translation difficulty: segment duration, speech rate, visual density in characters per second, lexical diversity, and estimated fingerspelling load. The analysis shows a corpus that is well matched to the 2–3 minute news-item format but strongly skewed across broadcasters and topics, with an open vocabulary and a non-trivial proportion of content requiring manual spelling. We discuss what these properties imply for the choice of modelling approach, and argue that gloss-free continuous translation, rather than the isolated recognition paradigm that dominates existing LSR work, is the only formulation compatible with data of this kind.

1 Introduction

Sign languages are full natural languages with their own phonology, morphology and syntax; they are not manual encodings of the surrounding spoken language (Yin et al., 2021). Romanian Sign Language is the primary language of the Deaf community in Romania, with an estimated 20,000–25,000 users. Access to information in LSR is mediated by a small number of human interpreters. Romanian audiovisual legislation obliges nationally broadcasting television services to provide sign language interpretation and synchronous subtitling for a minimum daily window of news and current affairs programming, and TVR Cluj became the first Romanian broadcaster to interpret its news bulletins in 2013.

Progress on automatic sign language translation has been driven almost entirely by data. The languages for which usable systems exist, German Sign Language, British Sign Language and American Sign Language, are precisely the languages for which large parallel corpora were released (Camgoz et al., 2018; Albanie et al., 2021; Shi et al., 2022). LSR has no such resource. The recently introduced RoCoISLR corpus (Rîpanu et al., 2025) is built from dictionary-style videos of *isolated* signs: roughly 9,000 clips covering approximately 6,000 glosses, with an average of well under two examples per gloss. The reported ceiling, a Top-1 accuracy of 34.1% with a Swin Transformer, is instructive: even on the comparatively easy isolated task, with clean dictionary-style input, LSR recognition is far from solved. To our knowledge, no continuous LSR translation corpus has previously been described. This design supports sign lookup and classification, but it cannot support translation of connected discourse, because connected signing is not a concatenation of citation forms. The first reason is co-articulation. In connected signing, adjacent signs influence one another's form: handshape, orientation and location may shift toward a neighbouring sign, and signs are produced faster and less fully articulated than in isolation (Neidle, 2023). A sign in running discourse therefore rarely matches its dictionary citation form, so a classifier trained on isolated, fully-articulated signs cannot be expected to recognise the same signs as they actually occur in continuous video.

This paper describes a corpus intended to fill this gap. Our contributions are:

1. We describe a pipeline for constructing a con-

tinuous LSR corpus from publicly available broadcast news, yielding approximately 902 hours of footage segmented into 16,638 topically labelled news items with aligned Romanian transcripts (section 2).

2. We provide a quantitative characterisation of the corpus along the axes that govern translation difficulty, namely duration, speech rate, visual density, lexical diversity and fingerspelling load, and relate each to a concrete modelling consequence (section 3).

3. We argue, on the basis of that characterisation, that the isolated recognition paradigm used in prior LSR work is not applicable here, and outline a gloss-free continuous translation formulation together with realistic expectations for what it can achieve at this data scale (section 4).

2 Corpus Construction

2.1 Source Selection

We collected publicly available news programming from the official YouTube channels of four Romanian broadcasters, chosen to span different editorial scopes:

- **TVR Iasi**, the regional studio of the Romanian public broadcaster serving the Moldavian region;
- **Prima TV**, a national commercial network;
- **Digi24**, a national rolling-news channel;
- **Euronews Romania**, the Romanian-language service of an internationally oriented news network.

The broadcasts carry a picture-in-picture LSR interpretation, provided under the accessibility obligations described above, occupying a fixed region of the frame. Videos were retained only where the interpreter inset is present for the full duration of the bulletin. Across the four sources the corpus involves [TODO: N] distinct interpreters. All footage is [TODO: resolution] at [TODO: frame rate] fps.

2.2 Segmentation into News Items

Broadcast bulletins are 20 to 40 minutes long and cover many unrelated stories. Treating a whole bulletin as a single translation unit is not viable: the sequence lengths are far beyond what current sequence-to-sequence models handle, and the resulting units would be topically incoherent. We therefore segment each bulletin into individual *news items*, self-contained stories on a single subject, bounded by anchor handovers and studio transitions. Segmentation was performed automatically using a large language model (OpenAI GPT-4o, accessed via the OpenAI API). The model was given the time-coded transcript of a full bulletin and prompted §A to identify topic boundaries. Segment boundaries returned by the model were mapped back onto the corresponding video time codes to extract each news item. Topic labelling of each resulting segment was performed separately, as described below.

2.3 Transcripts and Alignment

Each news item is paired with its Romanian transcript, obtained from YouTube automatic captions. Transcripts are time-coded and aligned to the *audio* track. This is a weak form of supervision for sign-to-text translation, because interpreters lag the speaker: the signing corresponding to a given subtitle typically begins some seconds after that subtitle appears, and the lag varies with syntactic complexity and interpreter strategy.

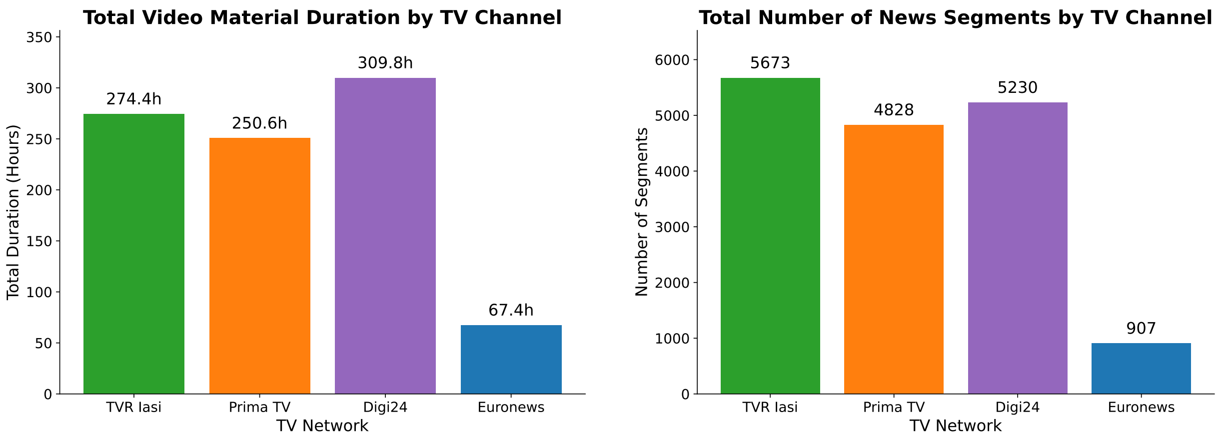
2.4 Topic Labelling

Each news item was assigned exactly one of nine topic labels (Justice, Culture, Healthcare, Economy, Education, Weather & Environment, Geopolitics, Politics, Sports). Labelling was carried out automatically using GPT-4o, accessed via the OpenAI API §B. The model was given the transcript of each news item together with the fixed set of nine topic labels and prompted to select the single best-fitting label per item. Topic labels serve two purposes: they permit the domain-restricted evaluation splits proposed in section 4, and they act as a proxy for lexical domain, which we show in subsection 3.6 to be a meaningful predictor of vocabulary diversity.

2.5 Derived Measures

We compute four derived measures over the transcript of each news item. Because the transcript records the spoken Romanian source, these measures characterise the *input* to the interpreting process and the *target* of an automatic sign-to-text system; they are proxies for, not direct measurements of, properties of the signing itself.

Speech rate (WPM). Word count divided by segment duration in minutes. This bounds the rate at



(a) Total duration of video material.

(b) Number of segmented news items.

Figure 1: Corpus scale by broadcaster. The two panels track one another only approximately: Digi24 supplies the most footage but not the most news items, because its segment length distribution has a heavier right tail.

which an interpreter must produce output and, correspondingly, the information density a translation model must recover per unit time.

Visual density (CPS). Character count divided by segment duration in seconds. CPS is the standard subtitling metric for reading load; the accepted accessibility ceiling for broadcast media is approximately 15 CPS.

Type-token ratio (TTR). Ratio of distinct lemmas to total tokens, computed with stop words removed. Stop word removal is essential here: Romanian is morphologically rich and function-word heavy, and including stop words compresses TTR towards a value that reflects segment length rather than lexical variety. TTR is length-sensitive, and our segments vary in length, so we interpret it comparatively across broadcasters and topics rather than as an absolute quantity.

Fingerspelling load. The proportion of tokens with no established lexical sign, which an interpreter must therefore render through the manual alphabet. We estimate this as the proportion of tokens that are proper nouns, acronyms, or out-of-vocabulary technical terms.

3 Corpus Analysis

3.1 Scale and Distribution across Broadcasters

The corpus comprises approximately 902 hours of broadcast footage. As Figure 1a shows, this material is distributed very unevenly across sources: Digi24 contributes 310 hours, TVR Iasi 274 hours and Prima TV 251 hours, together accounting for roughly 93% of the total, while Euronews Romania contributes 67 hours.

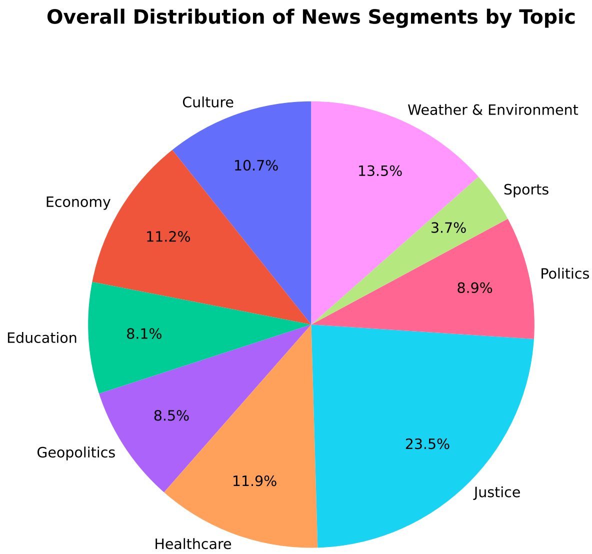


Figure 2: Distribution of news items by topic across the full corpus.

Segmentation yields 16,638 news items (Figure 1b): 5,673 from TVR Iasi, 5,230 from Digi24, 4,828 from Prima TV, and 907 from Euronews Romania. The two panels are broadly proportional but not exactly so. Digi24 supplies the largest volume of footage while ranking second in item count, and Euronews Romania yields the fewest items per hour of any source. Both facts follow from differences in typical item length, which we quantify directly in subsection 3.3.

3.2 Topic Distribution

Topic composition matters for a translation corpus because it determines which regions of the lexicon are covered. Figure 2 shows that the corpus

Corpus	Language	Hours	Signers	Vocab.	Type
UWB-07-SLR-P (Campr et al., 2008)	Czech SL	11.1	4	378	Isolated, lab
CISLR (Joshi et al., 2022)	Indian SL	—	71	4,765	Isolated, dictionary
RoCoISLR (Rîpanu et al., 2025)	Romanian SL	—	—	5,892	Isolated, dictionary
PHOENIX-2014T (Camgoz et al., 2018)	German SL	11	9	~3K	Continuous, broadcast
Content4All (Camgöz et al., 2021)	Swiss German / Flemish SL	253	—	—	Continuous, broadcast
BOBSL (Albanie et al., 2021)	British SL	1,447	37	77K	Continuous, broadcast
OpenASL (Shi et al., 2022)	American SL	288	~220	33K	Continuous, web video
YouTube-ASL (Uthus et al., 2023)	American SL	984	>2,519	60K	Continuous, web video
This work	**Romanian SL**	**902**	**[TODO]**	**[TODO]**	**Continuous, broadcast**

Table 1: Our corpus in the context of existing sign language resources. Dashes indicate quantities not reported in the corresponding paper or not applicable to the corpus type. Vocabulary figures are not directly comparable across isolated and continuous corpora: for isolated corpora they denote the number of gloss classes, for continuous corpora the number of distinct spoken-language word forms in the aligned text.

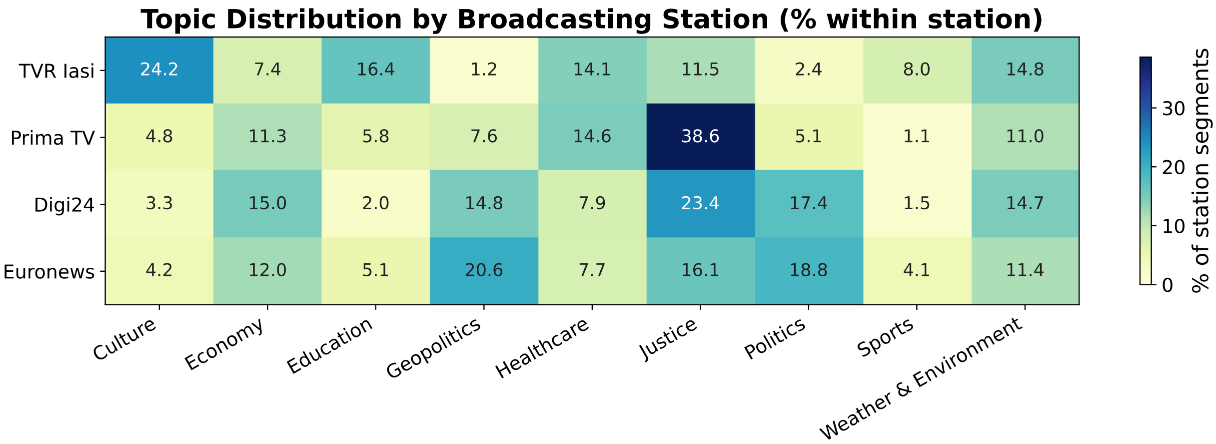


Figure 3: Topic distribution by broadcaster. The four networks show distinguishable editorial profiles, which induces a confound between broadcaster identity and lexical domain.

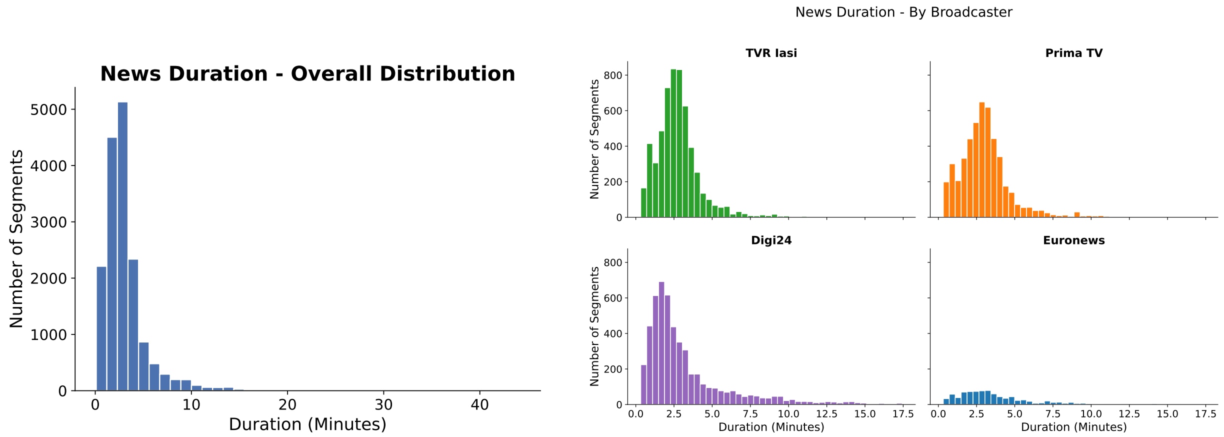
is skewed towards Justice (23.5%), Weather & Environment (13.5%) and Healthcare (11.9%), with Sports (3.7%) and Education (8.1%) at the opposite extreme. The remaining categories fall between 8% and 12%.

Figure 3 decomposes this by broadcaster, and the profiles align with each network's editorial remit. Prima TV is dominated by Justice, which alone accounts for 38.6% of its items, followed by Healthcare and Economy. Digi24 has the flattest profile of the four, spreading its items across Justice (23.4%), Politics (17.4%), Economy (15.0%), Geopolitics (14.8%) and Weather & Environment (14.7%), as one would expect of a rolling news channel. TVR Iasi shows the community-oriented profile characteristic of a regional studio, led by Culture (24.2%) and followed by Education, Weather & Environment and Healthcare, with Geopolitics and Politics almost absent at 1.2% and 2.4%. Euronews Romania, despite its smaller volume, concentrates on Geopolitics (20.6%), Politics (18.8%) and Justice (16.1%), consistent with its international orientation.

3.3 Segment Duration

Figure 4 shows that segment duration has a tight central tendency, with an overall median of 2.6 minutes and per-broadcaster medians ranging from 2.3 minutes for Digi24 to 3.2 minutes for Euronews Romania. This reflects a standardised news-item format rather than any property of our segmentation procedure.

The distribution has a long right tail, and its weight differs by source. Digi24 has the shortest median item yet the largest volume of footage, because its tail extends further than that of the other networks: a substantial number of its items run beyond ten minutes. These outliers are not segmentation errors but extended journalistic formats such as investigative reports, studio interviews and breaking news coverage. They matter operationally: a 2.6-minute segment at 25 fps is roughly 3,900 frames, already an order of magnitude longer than the clips typical of continuous sign language



(a) Full corpus.

(b) By broadcaster.

Figure 4: Distribution of news item duration. The median sits between 2.3 and 3.2 minutes across the four networks, with a long right tail of extended journalistic formats.

benchmarks, and the tail extends well beyond this.

3.4 Speech Rate

Speech rate is approximately normally distributed, with an overall median of 147 WPM (Figure 5). Segmenting by broadcaster (Figure 5b) leaves the shape of the densities essentially unchanged and shifts only their location. Three of the four networks are tightly grouped, with medians of 143 WPM for TVR Iasi, 142 WPM for Digi24 and 145 WPM for Euronews Romania. Prima TV is the single exception, sitting some 14 WPM higher at 157 WPM, which is consistent with the faster delivery of a commercial bulletin.

This near-consistency is convenient. It means that temporal properties of the input, that is how much content must be recovered per unit of video, are stable across most of the corpus, so a model need not adapt to source-specific pacing except in the case of Prima TV, and duration can be used as a reasonable proxy for content length during batching and curriculum design.

3.5 Visual Density and the Accessibility Ceiling

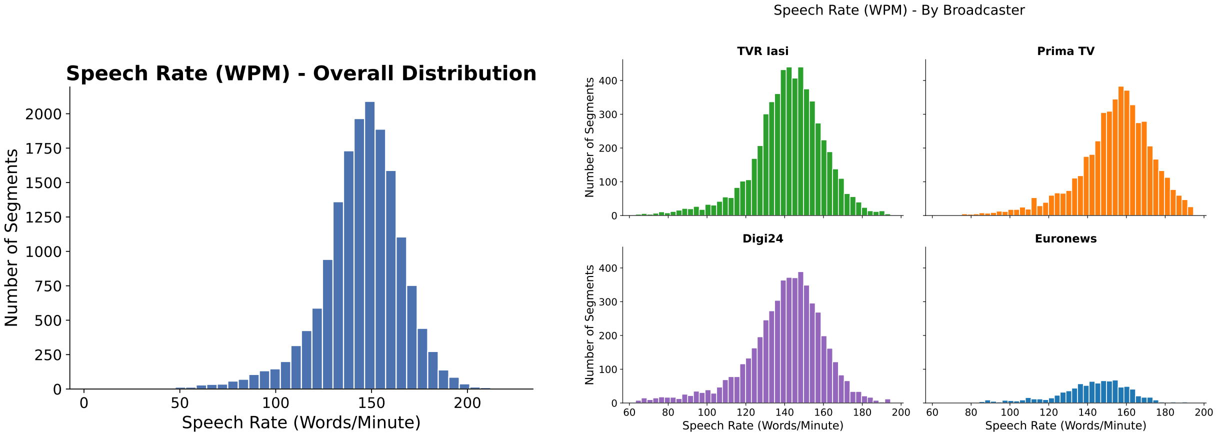
Characters per second is the standard measure of reading load in subtitling, with an accepted accessibility ceiling around 15 CPS. Above this rate, text transitions faster than a reader can comfortably process it.

Figure 6 shows an overall median of 14.2 CPS, just below the threshold: most news items are, in principle, paced for accessible presentation. The upper quartile and the outliers, however, exceed it. Broadcaster-level breakdown (Figure 6b) reveals differing profiles: Prima TV has the highest median at 14.9 CPS, clustering tightly against the limit, while Digi24 sits lowest at 13.7 CPS, with TVR Iasi at 14.2 and Euronews Romania at 14.0. All four sources carry a tail of segments approaching 20 CPS.

We note the limits of this measure. CPS quantifies the density of the Romanian text, not of the signing, and the two are related only indirectly: a competent interpreter facing a dense passage compresses and restructures rather than signing proportionally faster. The high-CPS tail is therefore best read as identifying segments where the interpretation is most likely to diverge structurally from the transcript, that is, where the weak alignment between video and text is weakest.

3.6 Lexical Diversity

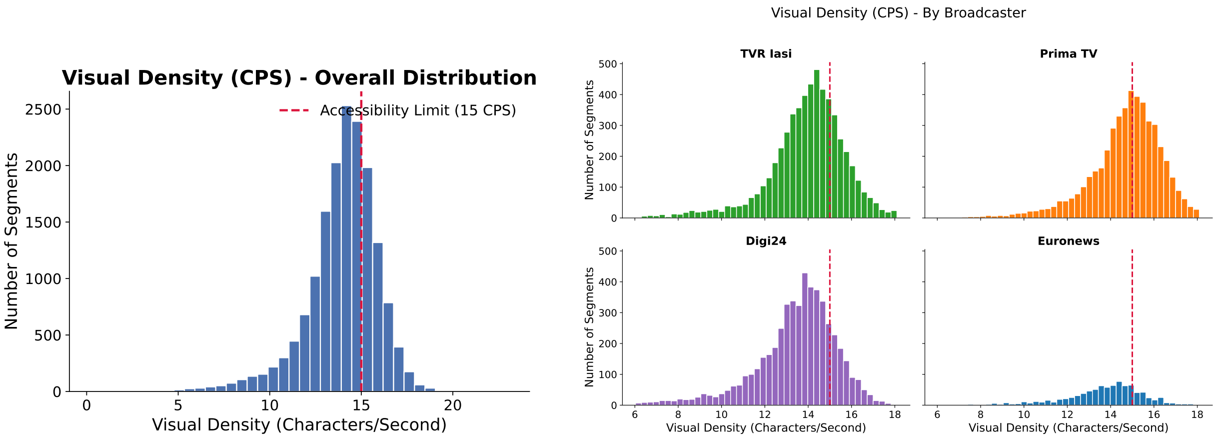
With stop words excluded, median TTR exceeds 0.7 for all four broadcasters (Figure 7a), ranging from 0.73 for Euronews Romania to 0.78 for TVR Iasi, with an overall median of 0.76. In a typical news item, then, more than seven in ten content words are unique. This is the single most consequential property of the corpus: it is an open-vocabulary domain, and it is not close to the closed-vocabulary regime in which sign language translation has historically succeeded. RWTH-PHOENIX-Weather 2014T, on which reported BLEU scores are highest, covers weather forecasts with a gloss vocabulary on the order of a thousand items and heavy formulaic repetition. Our corpus has no such ceiling. Digi24 is worth noting separately: its median is in line with the others, but its spread is the widest of the four, so a broadcaster-balanced split will not by



(a) Full corpus.

(b) By broadcaster.

Figure 5: Distribution of speech rate in words per minute. The four broadcaster densities overlap heavily and are centred on 140–160 WPM.



(a) Full corpus.

(b) By broadcaster.

Figure 6: Visual density in characters per second, against the 15 CPS accessibility threshold. The median sits just below the limit, but the upper quartile and outliers exceed it.

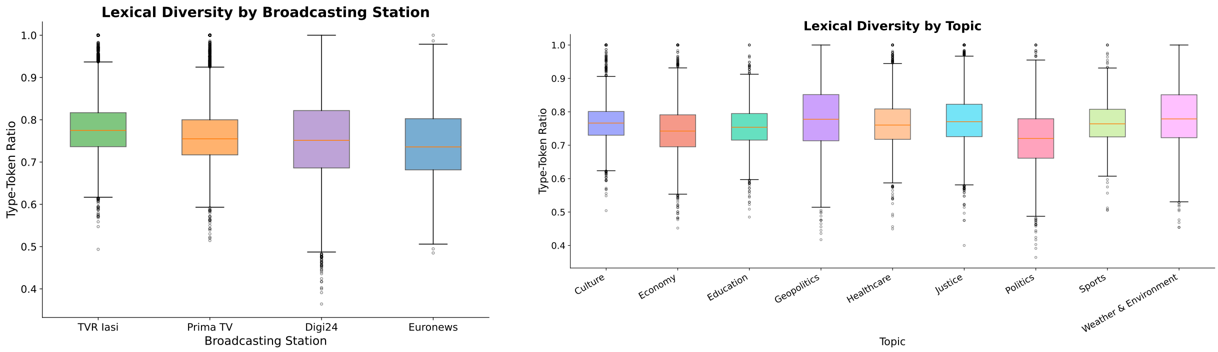
itself produce a diversity-balanced one.

The thematic breakdown (Figure 7b) shows systematic variation. Weather & Environment and Geopolitics have the highest median diversity, while Politics and Economy are lowest, reflecting the formulaic register of political and financial reporting. Since lexical diversity translates directly into the number of distinct signs a system must handle, this variation suggests a practical strategy: a domain-restricted subset drawn from the lower-diversity topics provides a tractable initial benchmark, against which the full open-domain corpus can be positioned as the harder target.

3.7 Fingerspelling Load

Estimated fingerspelling load, the proportion of content an interpreter must render through the manual alphabet, has an overall median of 6.1% and varies little across the four networks, from 5.3% for Prima TV to 6.6% for Digi24 (Figure 8). This appears to be a baseline rate for Romanian news discourse.

The tail is again the interesting part. High-load segments are those dense in proper nouns, foreign names and technical terminology without established signs. TVR Iasi carries the most extreme cases, with individual items above 40%. For a translation system this is the hardest category of content, for two compounding reasons. Fingerspelled sequences are visually rapid and low-amplitude, making them harder to recognise than lexical signs; and the target strings are precisely the named entities that a language model decoder cannot infer from context. Work on ASL has found that fingerspelling requires dedicated treatment within the translation pipeline rather than being absorbed by



(a) By broadcaster.

(b) By news topic.

Figure 7: Type-token ratio with stop words excluded. Median TTR exceeds 0.7 for all four broadcasters, indicating an open-vocabulary domain.

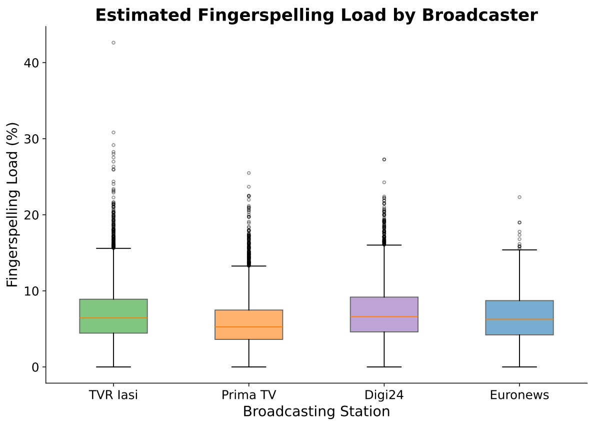


Figure 8: Estimated fingerspelling load by broadcaster.

a general model.

4 Modelling Considerations

The analysis above constrains the space of viable approaches more tightly than might be expected. We set out those constraints and the formulation they imply.

4.1 A Gloss-Free Formulation

The alternative is gloss-free translation, which maps sign video directly to spoken-language text without an intermediate symbolic layer, typically by coupling a visual encoder to a pretrained language model. Gloss-based translation, in which video is mapped to gloss and gloss to text, gives the best reported results on PHOENIX-2014T, but it requires gloss annotation of the training video. Producing such annotation is extraordinarily expensive: the Polish Sign Language Corpus required roughly twenty trained annotators over a decade to gloss 565 hours (Kuder et al., 2022). Gloss annotation of 902 hours of LSR is not available and will not become available within the scope of this work.

A pipeline consistent with our data has four stages:

1. Interpreter localisation. Crop the picture-in-picture region and track the interpreter. Because the inset position is fixed per broadcaster, this is largely a preprocessing step rather than a learned one.

2. Visual representation. Extract per-frame features, either as pose keypoints or as embeddings from a pretrained video encoder. Pose-based representations are compact, signer-invariant and cheap, an important consideration given that few interpreters appear in the corpus, at the cost of discarding appearance detail relevant to handshape and mouthing.

3. Temporal alignment. Recover the correspondence between signing and transcript. Since our transcripts are audio-aligned and interpreters lag the speaker, this cannot be assumed and should be estimated, following the sign-to-subtitle alignment literature (Bull et al., 2021).

4. Decoding. Generate Romanian text with a pretrained Romanian or multilingual language model conditioned on the encoded video.

4.2 Realistic Expectations

We state plainly what this corpus can and cannot be expected to support, since we regard mis-set expectations as the principal risk to work of this kind.

Open-domain sign language translation performs poorly in absolute terms. The high figures associated with PHOENIX-2014T reflect a closed weather-forecast domain of about a thousand glosses and do not generalise. Our corpus is open-domain by construction, with median TTR above 0.7, spanning nine topics from Justice to Sports. A single-digit BLEU-4 is the realistic expectation for an end-to-end system trained here, and this should be stated in advance rather than discovered.

Three further limits follow from the analysis. Signer diversity is low: BOBSL has 37 signers over 1,447 hours and YouTube-ASL more than 2,500, whereas broadcast interpretation in Romania is delivered by a small professional community, so a model risks fitting individual interpreters rather than LSR. Alignment is weak, and the high-CPS tail marks the segments where it is weakest. And named entities, at an estimated 6% fingerspelling load with a tail reaching far higher, will fail systematically.

5 Conclusion

We have described a continuous Romanian Sign Language corpus derived from broadcast news, approximately 902 hours segmented into 16,638 topically labelled news items with aligned Romanian transcripts, and analysed the properties that determine how it can be used. The corpus is well matched to a standardised 2–3 minute news format with stable delivery pace, but strongly imbalanced across broadcasters, open-vocabulary, and carrying a persistent fingerspelling load. Together these properties rule out the isolated recognition paradigm of prior LSR work and point towards gloss-free continuous translation, with correspondingly modest performance expectations. The resource is, to our knowledge, the first of its kind for LSR, and we hope it lowers the barrier to further work on a language that has so far been almost absent from the field.

Limitations

Several limitations should be weighed when interpreting these results.

The derived measures in subsection 2.5 are computed over Romanian transcripts, not over the signing. Speech rate, visual density, lexical diversity and fingerspelling load therefore characterise the source discourse and the load it places on an interpreter; they are not measurements of LSR. Fingerspelling load in particular is estimated from text using a lexical heuristic and has not been validated against annotated video, so the reported 6% figure should be read as an estimate of an upper bound on the relevant phenomenon rather than an observed rate.

Type-token ratio is sensitive to text length, and our segments vary substantially in length. The broadcaster and topic comparisons in subsection 3.6 are made across groups whose length distributions are similar, but the absolute values should not be compared to TTR figures from corpora with different segment lengths.

The corpus is drawn from four broadcasters, three of which supply 93% of the material, and is interpreted by a small number of professionals. It represents formal broadcast register and the signing of a small professional cohort; it does not represent conversational LSR, or the signing of Deaf people who are not professional interpreters. Models trained on it should not be assumed to transfer to everyday interaction.

Transcripts are aligned to audio rather than to signing, and we have not yet quantified the resulting temporal offset. Until that offset is measured, the effective quality of supervision the corpus provides remains uncertain.

Finally, broadcast interpretation is itself a translation, produced under real-time pressure with compression and omission. A system trained on this data learns to reproduce interpreted output, which is not identical to LSR as produced by Deaf signers in unconstrained settings.

Ethics Statement

The corpus is built from publicly broadcast material accessed through the broadcasters' official channels. Reuse for research is subject to the broadcasters' terms.

The interpreters appearing in the footage are identifiable individuals engaged in professional public-facing work. They have not consented to the use of their image as machine learning training data, and the granularity at which a sign language corpus captures a person, their signing style, facial expression and mouthing, exceeds what is normally at stake in the reuse of public footage. Prior sign language corpus projects have taken this seriously, adopting tiered access and withholding sensitive material rather than relying on anonymisation (Kuder et al., 2022).

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A Segmentation Prompt

The following system prompt was used with the OpenAI API (model GPT-4o) to segment each bulletin's time-coded transcript into individual news items, as described in §2.

```
You are an executive news producer.
Your task is to divide a time-coded
transcript into distinct news stories.

STRICT REQUIREMENTS:
Full time coverage. You may not leave
any gap between stories. The end_time
of one story must be the exact same
string as the start_time of the next
story.
Correct: Story 1 (00:00:00-00:04:21),
Story 2 (00:04:21-00:06:31).
Incorrect: Story 1 (00:00:00-00:04:21),
Story 2 (00:04:55-...).

Cover all text. A story includes
everything discussed under it,
interviews and on-location reports,
up to the next clear transition (e.g.
“moving on”, “next up”).

Respond exclusively in the following
JSON format:

{
  "stories": [
    {
      "story_id": 1,
      "short_topic": "Topic name",
      "start_time": "00:00:20",
      "end_time": "00:09:21"
    }
  ]
}
```

B Topic Labelling Prompt

The following system prompt was used with the OpenAI API (model GPT-4o) to assign a single topic label to each news item from the fixed set of nine categories, as described in §2.

You are a news editor. Classify the given text into STRICTLY ONE of the following categories: Politics, Economy, Education, Healthcare, Geopolitics, Justice, Sports, Culture, Weather & Environment.

Strict definitions and delimitation rules:
Politics: Political party activities, governmental decisions, elections, parliamentary activity, and central administration.
Economy: Finance, markets, business environment, inflation, taxes, budget, and macroeconomic indicators.
Education: Schools, universities, students, teachers, the Ministry of Education, and school reforms.
Healthcare: Hospitals, healthcare systems, diseases, vaccination campaigns, medicines, and public health policies.
Geopolitics: Wars, armed conflicts, major geopolitical tensions, international crises, and global strategic developments.
Justice: Courts of law, trials, crimes, criminal investigations, police, and legislation.
Sports: Sports competitions, national or club teams, athletes, and results from various disciplines.
Culture: Arts, films, theater, music, releases, cultural events, and entertainment news.
Weather & Environment: Weather forecasts, extreme weather phenomena, ecology, environmental protection, and climate change.

Respond ONLY with the exact name of the category from the list, without any other words, explanations, or punctuation.